2025 IEEE International Conference on Intelligent Rail Transportation (ICIRT)
October 11-12, 2025. Beijing, China

Cooperative Nonlinear Model Predictive Control for Multiple Locomotives of Heavy Haul Train on Long Steep Downhill Sections

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2025 IEEE International Conference on Intelligent Rail Transportation (ICIRT) | 979-8-3315-9751-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICIRT66379.2025.11216780

***Abstract*—This study develops a nonlinear model predictive control (NMPC) method to address cooperative control for multiple locomotives in a heavy haul train (HHT) operating on long steep downhill sections (LSDS). Firstly, a multi-point dynamic model for HHT is constructed by integrating electric braking force, air braking force, coupler force and running resistances. Secondly, by formulating a multi-objective cost function that integrates speed tracking accuracy, coupler force mitigation, energy efficiency, and stability constraints within the NMPC framework, this method achieves cooperative control of geographically distributed locomotives while explicitly addressing speed limits, coupler safety limits, and actuator constraints of the system. Finally, numerical simulations are conducted, and the results verify that the multiple locomotives cooperative controller can coordinated with air braking force to significantly improves speed tracking accuracy and reduces the amplitude of coupler forces, demonstrating enhanced operational safety capabilities for HHTs on LSDS.**

***Index Terms*—Heavy haul train, nonlinear model predictive control, electric braking force, air braking force.**

I. Introduction

With the increasing global freight demand, heavy haul trains (HHTs) play a key role in modern transportation by efficiently carrying large quantities of goods over long distances, especially for the transportation of bulk goods such as mineral resources and timber [1]. To enhance economic benefits, HHTs are developing towards being longer and heavier, with a length of several kilometers and a load capacity of tens of thousands of tons [2]. Moreover, in China's two major heavy-haul railway lines, long steep downhill sections (LSDS) constitute a substantial proportion, with downhill sections exceeding 12‰ slope spanning over 20 kilometers, necessitating the implementation of cyclic air braking strategies to maintain regulated speed limits [3]. This operational scale-up and complex operating conditions introduce enormous challenge for the heavy haul train (HHT) tracking control technology, particularly in coordinating multiple locomotives across such extended formations.

Target speed tracking control aims to propose a control algorithm with good stability, adaptive features and strong robustness for the characteristics of HHTs. At present, a wide range of advanced control methods have been developed for different problems of HHTs. He [4] developed a sliding mode controller with anti-saturation and anti-interference capabilities. Wang [5] considered the response process of control commands and the uncertainty of parameters, designed an adaptive nonsingular terminal sliding mode controller. Wang [6] designed a periodically intermittent cruise controller of HHT with uncertain parameters, and a robust and efficient cruise controller for HHTs based on state-dependent intermittent control was designed in [7]. Cai [8] studied the trajectory tracking problem for HHTs with time-varying reference speeds. Chen [9] proposed a HHT global tracking control scheme that can switch between the composite learning controller and the additional robust controller. In order to improve the control performance (energy efficiency, punctuality, or driving smoothness, etc.), model predictive control (MPC) has also been applied to the operation control of HHTs [10]. Wang [11] used the MPC method to study the robust optimal control problem under imperfect communication conditions. Considering the characteristic that the formation of HHTs is extremely long, both references [12] and [13] have studied the rationality of the "fence" method and applied it to the MPC algorithm.

This work was supported in part by the National Natural Science Foundation of China under Grant 52172322, Grant U22A2046, Grant 52272329, Grant U2468208, Grant 52372309, Grant U2468206, Grant U2469211, and Grant 52372310; in part by the Scientific Research Innovation Capability Support Project for Young Faculty under Grant 2025JBZD001 ; and in part by the Science and Technology Research Development Plan of China Railway Corporation under Grant L2023G004.

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In [14], a cascade control method based on MPC-PI control is proposed, which has better performance than using the two control methods alone. In [15], considering the three aspects of operational safety, service quality and energy consumption, a predictive control method for HHT models considering the electronically controlled pneumatic (ECP) braking system was developed. Dong [16] designed the fuzzy predictive controller in neutral sections and steep slopes.

The multi-point model is mostly adopted in the study of HHTs. However, many studies oversimplify HHT dynamics by treating locomotives and wagons as a single unit, ignoring coupler forces between vehicles [5], [8], [10]. Furthermore, many control methods only focus on analyzing train operations under moderate gradient conditions that do not require air braking [4], [6], [7], [11], failing to address the complex dynamics encountered during steep descents where gravitational acceleration exceeds available braking capacities. In the existing studies considering air braking, most of the models employ the ECP system which can instantaneously respond to air braking commands [15]. By contrast, China adopts the locomotive control (LOCOTROL) system. In this system, the braking signal is conveyed from the locomotive to the wagons via the air wave within the train tube, resulting in a non-negligible time delay. This inherent characteristic fundamentally constrains braking synchronization of HHTs during extended descents, constituting a persistent research bottleneck [17]. Conventional control methodologies relying heavily on driver experience demonstrate limitations in coordinating multi-locomotive collaboration, particularly when locomotives that are located at different positions output the same control force. Such approaches frequently lead to speed limit violations, abrupt coupler force fluctuations, or unintended stops for air replenishment, compromising operational efficiency and potentially posing a hazard to safety [18].

Motivated by the aforementioned discussions, this paper develops a nonlinear model predictive control (NMPC) algorithm to control the output electric braking force of the distributed locomotive, which coordinates with the air braking for speed regulation to track the preset speed. The contributions of this paper include:

- A decentralized model predictive controller has been designed for the operation scenario of HHTs on LSDS. The controller dynamically adjusts individual braking outputs according to real-time states, coordinates electric braking forces with cyclic air braking sequences, ensuring accurate tracking of the predefined speed trajectory while minimizing coupler forces and reducing energy consumption.
- In contrast to conventional models that neglect pneumatic brake wave propagation dynamics, this work explicitly integrates air braking delays into the nonlinear train dynamics. This advancement enables accurate characterization of distributed braking coordination challenges on LSDS, extending HHT control strategies beyond traditional flat-terrain scenarios.

The remainder of this paper is organized as follows: Section II presents the train dynamics modeland elaborates the control objectives. Section III details the design of the nonlinear model predictive controller. Section IV validates the proposed approach through simulation studies, demonstrating its correctness and effectiveness. Finally, Section V summarizes the research findings and discusses future research directions.

II. Problem Description

A. The Dynamic Model of Heavy Haul Train

Combined HHTs usually consist of multiple locomotives and a large number of wagons, as shown in Fig. 1. Suppose that there are n locomotives and $(N-n)$ wagons in a HHT. Define the position sequence number of the locomotive as the set $\mathscr{L} = \{i | i = j_1, j_2, \cdots, j_n\}$ and the position sequence number of the wagons as the set $\mathscr{N} = \{i | i = 2, 3, \cdots, N \wedge i \notin \mathscr{L}\}$. The longitudinal dynamics of HHTs manifest through complex force interactions: each vehicle unit experiences coupler forces, running resistance and air braking force. In addition, the locomotives will output electric traction/braking force. Regard each locomotive or wagon as a separate point, then, the longitudinal dynamics equation of the HHT can be represented as

$$\begin{aligned} &\dot{s}_i(t) = v_i(t) \\ &m_i \dot{v}_i(t) = u_i^e(t) - u_i^a(t) + f_{i-1}^c(t) - f_i^c(t) - R_i(t), i \in \mathscr{L} \\ &m_i \dot{v}_i(t) = -u_i^a(t - \tau_i) + f_{i-1}^c(t) - f_i^c(t) - R_i(t), i \in \mathscr{N} \end{aligned} \tag{1}$$

In (1), m_i, p_i, and v_i are the mass, position, and velocity of the ith particle, respectively. The control force output by the locomotive is expressed as u_i.

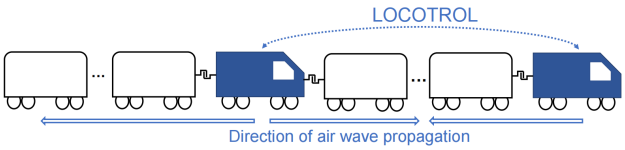


Fig. 1. Schematic diagram of the combined heavy haul train model.

The air braking system slows trains by directing compressed air from reservoirs into brake cylinders, forcing brake shoes against wheels to convert kinetic energy into heat through friction, and the air braking force u_i can be calculated by

$$u_i = \vartheta_b \cdot \varphi_b \cdot \beta_s \cdot 10^3, \tag{2}$$

where ϑ_b and φ_b are the equivalent emergency braking ratio and the equivalent friction coefficient, respectively, β_s denotes the service braking coefficient. In LSDS, air braking cannot be applied continuously to prevent brake failure. After each application, sufficient time (typically 180s) is required to recharge the air reservoirs before subsequent braking, so the air braking force must be implemented and released in stages. What's more, due to the physical characteristics of china's air braking system, the control signals for air braking force application or release are simultaneously transmitted from all locomotives to the wagons on both sides through the brake

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tubes, as shown in Fig.1. Therefore, the delay time of the ith wagon is given by

$$\tau_{i,1} = \frac{L_i}{v_{wave2}}, \tau_{i,2} = \frac{L_i}{v_{wave2}}$$

where L_i is the distance between the ith wagon and the nearest locomotive , v_{wave1} and v_{wave2} are respectively the propagation speeds of the braking wave and the relief wave.

The couplers connecting the carriages can be approximated as a "spring-damper" system, so the in-train force can be established as follows:

$$f_i^c = k(s_i - s_{i+1} - l_0) + d(v_i - v_{i+1}) \tag{3}$$

where f_i^c is the rear coupler force of the ith carriage, l_0 is the centroid distance between adjacent carriages when there is no relative displacement, k_i is the stiffness coefficient and d_i is the damping constant, which are determined by the characteristics of couplers in the HHT.

The running resistance R_i can be divided into two parts: the basic resistance R_i^b and the additional resistance R_i^a. Generally speaking, the basic resistance mainly includes rolling resistance and aerodynamic resistance, depending on the HHT's traveling speed and its physical properties. Empirically, the Davis equation can be used to approximate the calculation of fundamental resistance [12]:

$$R_i^b = m_i(c_0 + c_1 v_i + c_2 {v_i}^2). \tag{4}$$

Additional resistance includes ramp resistance and curve resistance:

$$\begin{aligned} R_i^a &= R_i^g + R_i^c \\ R_i^g &= m_i g \sin\theta_i \\ R_i^c &= m_i g \tfrac{700}{R} \end{aligned} \tag{5}$$

where R is the track curve radius and θ_i is measured in meters per thousand.

Choosing $x = [s_1, s_2, \ldots, s_N, v_1, v_2, \ldots, v_N]^T$ as the state variable, and $u = [u_1^e, u_2^e, \ldots, u_{j_n}^e]$ as the decision vector (control variable). Since during operation, the time points for the application and relief of air braking are pre-set based on geographical location and other factors, and the delay time of air braking for each carriage remains constant, therefore, (1) can be expressed in the form of the following nonlinear function:

$$\dot{x} = f'(x, u). \tag{6}$$

Then, the zero-order hold method with sampling period T_s is adopted to discretize this model, obtaining:

$$x(k+1) = f(x(k), u(k)). \tag{7}$$

This discrete model is then used in the following controller design in a NMPC framework.

B. Problem Formulation

Combined HHTs exhibit distinctive dynamic characteristics due to their extended configuration. Typically, in a combined HHT, locomotives are distributed at different positionsit is obviously unreasonable for locomotives to output the same control force. Therefore, the objective of this paper is to control the electrical braking force of the locomotive when the velocity-time curve and the air braking strategy have been determined, cooperate with the air braking for speed regulation, ensure that the train can track the preset speed curve, implement air braking control or air supply mitigation at the set speed in the correct displacement, thereby preventing hazardous scenarios (e.g., coupler overloads or unintended stops) while minimizing in-train forces and energy consumption.

III. NMPC's Application To Heavy Haul Trains

NMPC is an advanced control methodology that iteratively optimizes system performance by solving a constrained finite-horizon optimization problem based on a dynamic process model. The core idea of the algorithm is at each control step k, using the current state $x(k)$ and a nonlinear dynamical model, under the condition of satisfying the constraints, it computes a sequence of optimal control actions

$$U^*(k) = \left[\ u^*(k|k), u^*(k+1|k), \cdots, u^*(k+N_c-1|k)\ \right]^T$$

by minimizing a predefined cost function J. Here, N_c represents the control horizon, and the method simultaneously predicts future system states over a longer prediction horizon N_p. When $N_c < N_p$, the controller assumes constant control inputs beyond the control horizon (i.e.,$u(k+i|k) = u(k+N_c-1|k)$ for $i \geq N_c$). To address practical challenges such as model inaccuracies and external disturbances, only the first control action $u^*(k|k)$ from the optimized sequence $U^*(k)$ is applied to the system. The entire process is then repeated at the subsequent time step $k+1$, following the receding horizon principle where the optimization window advances while maintaining fixed horizon lengths N_c and N_p.

By unifying prediction, optimization, and feedback mechanisms, this approach is particularly advantageous for complex systems like HHTs due to its inherent ability to handle nonlinear dynamics, explicitly incorporate physical and safety constraints, and dynamically balance competing objectives.

A. Cost Function Design

As clarified in Section II, the control of HHTs necessitates simultaneous optimization of three critical objectives: speed tracking accuracy, coupler force minimization, and energy efficiency. Consequently, the objective function integrates these three priorities through the following components:

(1) Speed tracking: To minimize deviation from the reference state $x_r = [s_{ref}, v_{ref}]^T$, a quadratic penalty on state deviations is imposed:

$$J_1 = \sum_{j=1}^{N_p-1} \|x(k+j|k) - x_r(k+j)\|_{Q_1}^2. \tag{8}$$

where $Q_1 \in \mathbb{R}^{2N\times 2N}$ is a block-diagonal weighting matrix structured as: $Q_1 = blkdiag(Q_s \cdot I_N, Q_v \cdot I_N)$. Here, Q_s and Q_v penalize positional and velocity tracking errors, respectively.

(2)Coupler force minimization: To mitigate longitudinal impulse and prevent mechanical failures, coupler forces between adjacent vehicles need to be minimized. As shown

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in (2), coupler forces depend on relative displacements and velocities, leading to the reformulated cost:

$$J_2 = \sum_{j=1}^{N_p-1} \|M(x(k+j|k) - x_r(k+j))\|_{Q_2} \tag{9}$$

where the transformation matrix $M \in \mathbb{R}^{(2N-2)\times 2N}$ extracts relative dynamics between consecutive vehicles:

$$M = \begin{bmatrix} M' & \\ & M' \end{bmatrix}, M' = \begin{bmatrix} 1 & -1 & & \\ & 1 & -1 & \\ & & \vdots & \\ & & 1 & -1 \end{bmatrix}_{(N-1)\times N}$$

Similar to Q_1, $Q_2 \in \mathbb{R}^{(2N-1)\times(2N-1)}$ consists of two parts: the weight coefficient of the relative displacement Q_{ds} and the relative velocity difference Q_{ds}, and its form is: $Q_2 = blkdiag(Q_{ds} \cdot I_{N-1}, Q_{dv} \cdot I_{N-1})$.

(3) Energy efficiency: Penalize excessive use of electric braking to reduce energy consumption:

$$J_3 = \sum_{j=1}^{N_p-1} \|u_i(k+j|k)\|_R^2, \tag{10}$$

where $R \in \mathbb{R}^{n\times n}$ is the weight coefficient of the electric braking force.

What's more, to ensure closed-loop stability of the proposed NMPC framework, a terminal cost function $V_f(x(k+N_p|k))$ and terminal constraint set Ω are explicitly designed based on Lyapunov theory. The terminal cost function is defined as a quadratic penalty on the terminal state deviation from the reference trajectory:

$$V_f(x(k+N_p|k)) = \left\|x(k+N_p|k) - x_r(k+N_p)\right\|_P^2. \tag{11}$$

where $P \in \mathbb{R}^{2N\times 2N}$ is a positive-definite weighting matrix. To guarantee stability, P is computed offline by solving the Discrete algebraic riccati equation (DARE) derived from the linearized system dynamics around the reference trajectory.

The terminal set Ω is defined as a positively invariant set where a local stabilizing controller $u = \kappa(x)$ satisfies:

$$\begin{cases} x \in \Omega \Rightarrow u = \kappa(x) \in \mathscr{U} \\ f(x, \kappa(x)) \in \Omega \\ V_f(f(x,\kappa(x)) \le V_f(x) - \|x - x_r\|_Q^2 - \|\kappa(x)\|_R^2. \end{cases} \tag{12}$$

For computational tractability, Ω is approximated as an ellipsoidal set:

$$\Omega = \left\{ x \in \chi \,|\, \|x - x_r\|_Q^2 \le \gamma \right\} \tag{13}$$

where $\gamma > 0$ is a scalar determined via offline optimization to maximize the region of attraction while ensuring constraint satisfaction.

In conclusion, the final design of the cost function is as follows:

$$\begin{aligned} J &= J_1 + J_2 + J_3 + V_f \\ &= \sum_{j=1}^{N_p-1} (\|x(k+j|k) - x_r(k+j)\|_{Q_1}^2 \\ &\quad + \|M(x(k+j|k) - x_r(k+j))\|_{Q_2}^2 + \|U(k+j|k)\|_R^2) \\ &\quad + \left\|x(k+N_p|k) - x_r(k+N_p)\right\|_P^2 \\ &= \sum_{j=1}^{N_p-1} \left(\|x(k+j|k) - x_r(k+j)\|_Q^2 + \|u(k+j|k)\|_R^2 \right) \\ &\quad + \left\|x(k+N_p|k) - x_r(k+N_p)\right\|_P^2 \end{aligned} \tag{14}$$

where $Q = Q_1 + M^T Q_2 M$.

B. Operational Constraints

The control system must operate within a set of physical, operational, and safety limitations inherent to HHTs. These constraints arise from the hybrid nature of electric braking systems, coupler strength thresholds and track speed regulations. Specifically, the controller must ensure:

(1) Actuation limits for electric braking forces, dictated by locomotive capabilities :

$$\begin{aligned} &u_{\min} \le {u_i}^e \le 0 \\ &-\Delta u_{\max} \le \Delta {u_i}^e \le \Delta u_{\max}, i \in \mathscr{L}. \end{aligned}$$

(2) Safety-critical bounds on coupler forces to prevent structural failure or derailment:

$$f_i^c \le f_{\max}^c, i = 1, 2 \ldots N-1.$$

(3) Relative displacement limits between adjacent vehicles to prevent excessive coupler elongation:

$$|s_{i+1} - s_i| < \Delta s_{\max}, i = 1, 2 \ldots N-1.$$

(4) Speed limits to avoid overspeeding and hook breakage during low-speed air brake relief:

$$v_{\min} \le v_i \le v_{\max}, \forall i.$$

Define the constraints regarding the system state as the set $\mathscr{X}$ and the constraints regarding the system control force as the set $\mathscr{U}$.

C. NMPC Algorithm

Based on the formulated train dynamics model, cost function, and system constraints, the multiple locomotives NMPC algorithm for a HHT can be formally defined as solving the following constrained optimization at each discrete time instant k:

$$\begin{aligned} &\min_U J(x_0, U, N_p, N_c) \\ &s.t. (\forall j = 0, 1, 2, \ldots N_p - 1) \\ &\begin{cases} x(k+j+1|k) = f(x(k+j|k), u(k+j|k)) \\ x(k+j|k) \in \chi \\ u(k+j|k) \in \mathscr{U} \\ x(k|k) = x(k) = x_0 \\ x(k+N_p|k) \in \Omega \subseteq \chi \end{cases} \end{aligned} \tag{15}$$

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The optimal control sequence $U^*(k) = [u^*(k|k), u^*(k+1|k), \cdots u^*(k+N_c-1|k)]^T$ obtained by solving the optimization problem comprises N_c control vectors. Following the receding horizon principle of NMPC, only the first control input $u^*(k|k)$ is applied to the system to update the train state at time k, yielding $x(k+1) = f(x(k), u^*(k|k))$. This updated state $x(k+1)$ then serves as the initial condition for the subsequent optimization at time $k+1$, thereby enabling closed-loop real-time control and dynamic adaptation to evolving operational conditions.

The NMPC algorithm process can be summarized as follows:

Algorithm 1 NMPC Algorithm for HHT

Step 1. Initialize controller parameters: prediction horizon N_p, control horizon N_c, and select weighting matrices Q, R, calculate the matrix P and parameter γ offline;

Step 2. Obtain the initial system state $x(k)$ at the current instant k;

Step 3. Formulate optimization problem 1 based on the prediction model, operational constraints, and the performance index function defined in (14);

Step 4. Solve the optimization problem to obtain the optimal control sequence U^*, then apply the first element $u(k) = u^*(k|k)$ to the system;

Step 5. At the next time instant, repeat Steps 2 to 5 to iteratively update the control process.

IV. SIMULATION

In this section, the simulation is developed in MATLAB to validate the efficacy and operational stability of the proposed NMPC method for multiple locomotives operation of a HHT. The simulation object is a 20,000-ton HHT on the Shuohuang Railway with a "1+1" formation (1 HXD1 locomotive + 108 C80 wagons + 1 HXD1 locomotive + 108 C80 wagons). Table I lists the parameters of the train.

TABLE I
ASSUMED VALUE OF RELEVANT VARIABLE PARAMETERS

Parameter	Value	Parameter	Value
HXD1 mass	$200t$	c_0	$7.6658 \times 10^{-3} Nkg^{-1}$
C80 mass	$100t$	c_1	$1.08 \times 10^{-4} Ns(mkg)^{-1}$
HXD1 length	$26m$	c_2	$2.06 \times 10^{-5} Ns^2(m^2kg)^{-1}$
C80 legnth	$12m$	k	$10 \times 10^7 Nm^{-1}$
Maximum electric braking force	$-461kN$	d	$10 \times 10^5 Nsm^{-1}$

To reduce computational complexity and enhance controller real-time performance, the fence method [13] is implemented in the model prediction process. Specifically, every 108 wagons are divided into four aggregated mass points through geometric partitioning, resulting in a total of 10 mass points for the entire 216-vehicle formation. The air braking delay time of each aggregated mass point is determined by the delay at the geometric center of its constituent wagons.

We selected the velocity-time curve generated in [18] as the target curve for tracking. Select $N_c = N_p = 6, T_S = 0.5s$, $Q_s = 0$, $Q_v = 1000$, $Q_{ds} = 10$, $Q_{dv} = 10$, $R = diag(1 \times 10^{-4}, 1 \times 10^{-4})$. The simulation results are shown in the following figures.

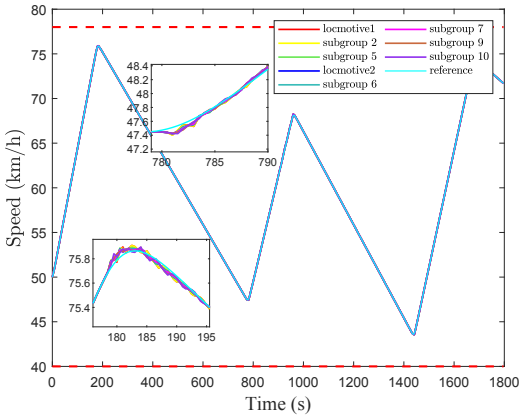


Fig. 2. Velocity tracking results.

Fig.2 displays the speed tracking performance of locomotives and selected train subgroups, while Fig.3 shows the speed errors of the two locomotives. It can be seen that the speed will exhibits noticeable fluctuations during air braking phase transitions (application/release). However, under the controller's action, the locomotive speed error (below 0.06 m/s) stabilizes within a specific time period, ultimately converging to $10^{-4} m/s$.

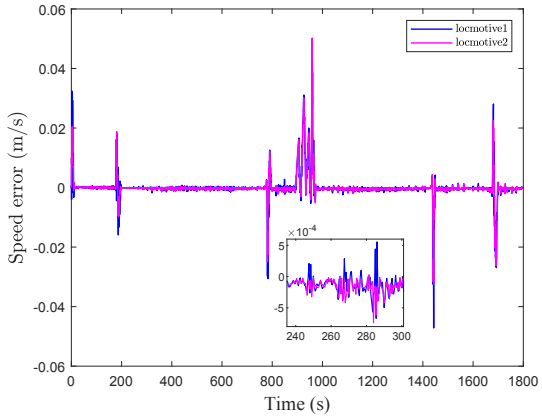


Fig. 3. Speed error of the locomotives.

Fig.4 presents the optimized electric braking forces output by locomotives, with all values remaining within actuator saturation limits ($-460kN$), and it can be seen that the control force output by locomotive 2 is greater.

Fig.5 shows the dynamic variation of the coupler forces between various particles of the train over time. An analysis reveals that throughout the entire operation process, the coupler forces of the train remain consistently within the safe threshold range($1600kN$). It is noteworthy that the peak values of the coupler forces mainly occur during the transition

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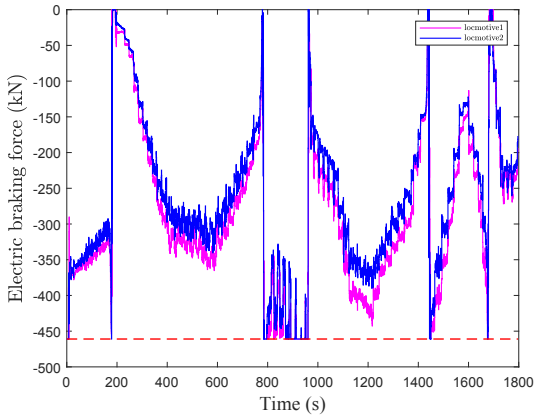


Fig. 4. Electric braking force of locomotives.

of the air braking condition. A further observation shows that the coupler forces near the locomotives are significantly higher. Especially notable is that the locomotive 2 which is located in the middle of the train, as it needs to undertake the dual traction tasks of "pulling the front and dragging the rear" simultaneously, results in the maximum coupler forces at both its front and rear couplers.

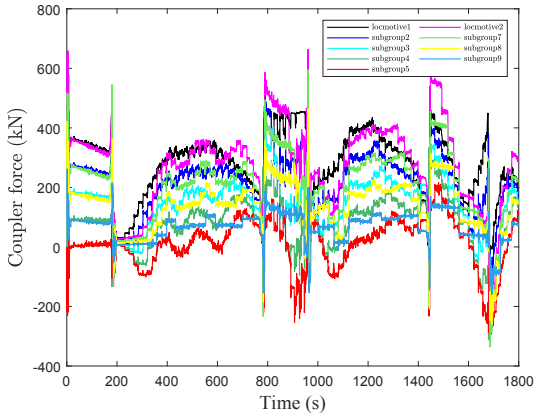


Fig. 5. The coupler forces in the HHT.

V. Conclusion

This study proposes a NMPC method for coordinating multiple locomotives in a HHT operating on LSDS to accurately track the speed trajectory. The proposed method takes speed tracking accuracy, coupler force mitigation, and energy efficiency as control objectives, while simultaneously considering the safety constraints of HHT operations. Finally, a simulation experiment is conducted on LSDS. The results show that the maximum inter-locomotive speed difference is about 0.05 m/s, and the maximum coupler force is approximately 640 kN, which is much lower than the coupler limit, demonstrating the effectiveness of the proposed method. Future work will extend to stochastic adhesion conditions, hardware-in-loop validation, and multi-train corridor optimization, advancing intelligent rail freight systems.

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